Runge - Kutta Method (R - k)

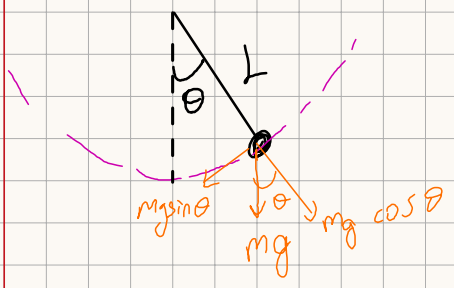
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Differential Equation Solver Simulator

- 48 methods are developed based on Runge - Kutta
- The simulator should present different methods to users In that way user can make comparative analysis

Simple Pendulum

Numerical Solution of Simple Pendulum



Sol

$$m\vec{a} = -mg\sin\theta$$

$$L\frac{d^2\theta}{dt^2} = -g\sin\theta$$

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0$$

Small angle approximation:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0$$

① Let's solve this with R-K here.

R-K solves $y' = f(x, y(x))$ which is a first-order differential equation

② So we need to make order reduction here at first to use R-K method

$$x_1 = \theta \qquad \omega = \sqrt{\frac{g}{L}}$$

$$x_2 = \frac{d\theta}{dt} = \dot{x}_1$$

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0 \longrightarrow \dot{x}_2 + \omega^2 x_1 = 0$$

$$\boxed{\dot{x}_2 = -\omega^2 x_1}$$

Matrix Notation

$$x_1 = \theta \qquad \dot{x}_1 = x_2$$

$$x_2 = \dot{\theta} \xrightarrow{\frac{d}{dt}} \dot{x}_2 = -\omega_2 x_1$$

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -\omega^2 x_1 \end{bmatrix}$$

Taylor Series

To obtain 2nd ODE via R-K we use Taylor series expansion ∴

$$y(x+h) = y(x) + h\,y'(x) + \frac{h^2}{2}y''(x) + O(h^3)$$

* Substitude $y'(x) = f(x, y(x))$

$$y'' = \frac{d}{dx}y'(x) = \frac{d}{dx}f(x, y(x)) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial x}$$

* we defined $y''(x)$ here we can substitute it in Taylor expansion

$$y''(x) = f_x(x,y) + f_y(x,y)\,f(x,y)$$

* Revised Taylor expansion.

$$y(x+h) = y(x) + h\,f(x, y(x)) + \frac{h^2}{2}\left[f_x(x, y(x)) + f_y(x, y(x))\,f(x, y(x))\right] + O(h^3)$$

$$f(x+h, y+k) = f(x,y) + h\,f_x(x,y) + k\,f_y(x,y) + \cdots \quad (1)$$

take $\rightarrow k = h\,f(x,y)$

$$f(x+h, y+k) = f(x,y) + h\,f_x(x,y) + h\,f_y(x,y)\,f(x,y) + O(h^2) \quad (2)$$

* Now it is similar to here

Substitude (2)

Edit the Taylor Series

$$\Rightarrow y(x+h) = y(x) + \frac{h}{2} f(x,y) + \frac{h}{2}\left[f(x,y) + h\, f_x(x,y) + h \cdot f_y(x,y)\, f(x,y)\right] + O(h^3)$$

$$y(x+h) = y(x) + \frac{h}{2} f(x,y) + \frac{h}{2} f\left(x+h,\ y + \underbrace{h \cdot f(x,y)}_{k}\right) + O(h^3)$$

* Substitute:

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x+h, y+k) = f(x+h,\ y + h\, f(x,y))$$

$$k_2 = f(x_i + h,\ y_i + h\, k_1)$$

* The Final Equation

$$y(x+h) = y(x) + \frac{h}{2} k_1 + \frac{h}{2} k_2$$

$$\Downarrow$$

$$y_{i+1} = y_i + \frac{h}{2}(k_1 + k_2)$$

* We explain teorical part until here

* How to transfer it to code?

Coding Part

① First of all we define our vector

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta \rightarrow \begin{cases} \frac{d\theta}{dt} = w \\ \frac{dw}{dt} = \frac{-g}{L}\theta \end{cases}$$

② So the state vector.

$$\textcircled{1}\ y_i = \begin{bmatrix} \theta_i \\ w_i \end{bmatrix} \xrightarrow{d/dt} \textcircled{2}\ f(x_i, y_i) = \begin{bmatrix} w_i \\ -\frac{g}{L}\theta_i \end{bmatrix} \quad (k_{1,\theta},\ k_{1,w})$$

③ Initial slopes:

$\textcircled{1}\ k_{1,\theta} = w_i$

$\textcircled{2}\ k_{1,w} = -\frac{g}{L}\theta_i$

④ Predicted secondary slopes (Euler method)

$\textcircled{3}\ \theta_2 = \theta_i + k_{1,\theta} \cdot h \rightarrow \textcircled{5}\ k_{2,\theta} = w_2$

$\textcircled{4}\ w_2 = w_i + k_{2,w} \cdot h \rightarrow \textcircled{6}\ k_{2,w} = -\frac{g}{L}\theta_2$

Improved Euler method:

$$y_{i+1} = y_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h$$

$$\textcircled{7}\ \theta_{i+1} = \theta_i + \left(\frac{1}{2}k_{1,\theta} + \frac{1}{2}k_{2,\theta}\right)h$$

$$\textcircled{8}\ w_{i+1} = w_i + \left(\frac{1}{2}k_{1,w} + \frac{1}{2}k_{2,w}\right)h$$

⑨ Define in code → graph it.